Data Warehouse and Pipeline

Week 2: Data Warehouse Overview

Keeyong Han

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9. Example Data System
10. Lab1/Demo/Homework

Recap of 1st Week

Data Flow

Site traffic &
3rd party data

1

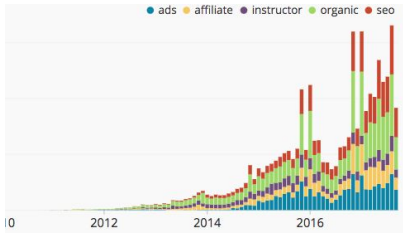
ELT

ETL

Data warehouse

Data Team

2



Business
Insights

3

Site Services

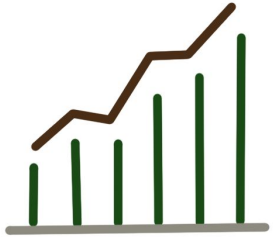
Scalable Product Improvement

Data System Evolution

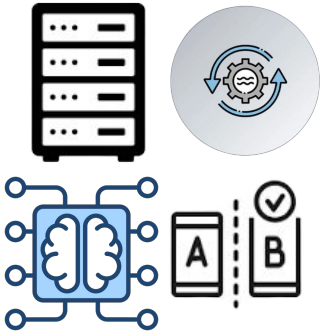
Data Sources

Data Pipelines

Data Storages



BI Tools



Study Materials

- You can read these from oreilly.com for free:

1. AI Engineering: Building Applications with Foundation Models, Chip Huyen
2. Designing Machine-Learning Systems, Chip Huyen
3. Designing Data-Intensive Applications 2nd edition: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems, Martin Kleppmann.
4. The Data Warehouse Toolkit, 3rd Edition: The Definitive Guide to Dimensional Modeling, Ralph Kimball and Margy Ross.

- SQL: https://leetcode.com/studyplan/top-sql-50/
- Python: https://www.udemy.com/course/100-days-of-code/

Office Hours & Classroom & Snowflake Academia

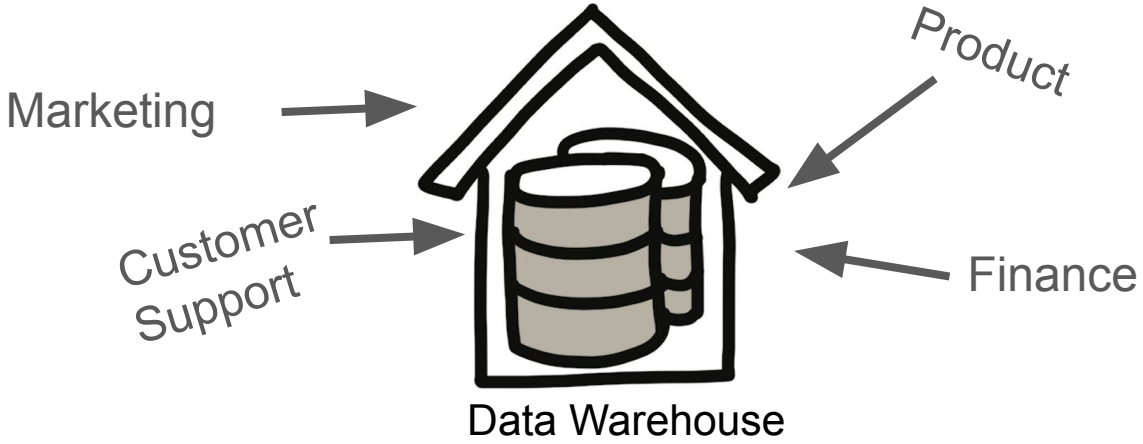
- Instructor: Keeyong Han (keeyong.han@sjsu.edu):
  - 2 to 3PM Tuesday at CL 418A
- Section 24: 3:00pm to 4:00pm, every Friday (Jeff)
- Section 23: 4:30pm to 5:30pm, every Thursday (Pranav)

- For Snowflake Academia, make sure to store your password :)
- Also if you haven’t got any email about the account, let me know

Definition of Data Warehouse

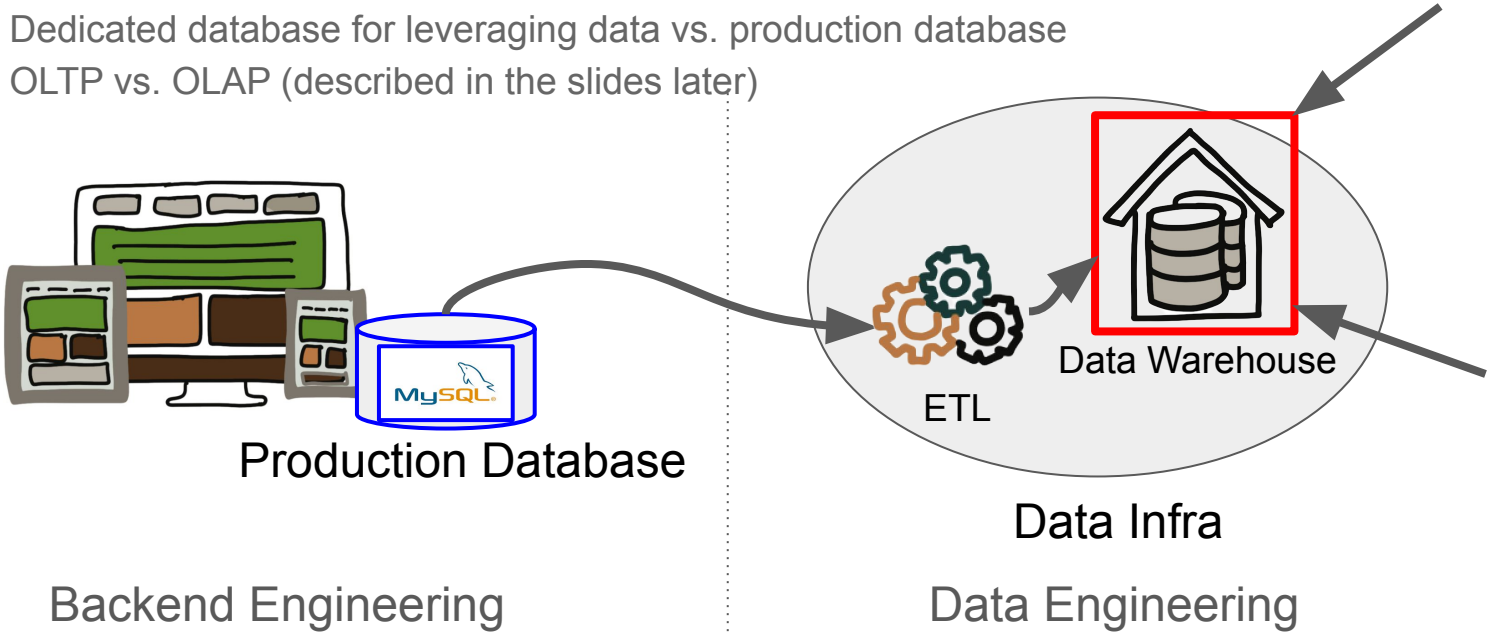
Data Warehouse: Central Data Storage of your Company

- Central Database with all the data needed for the company
  - Stores all kinds of raw data from different sources
  - Still SQL based relational database (data modeling still matters)
- Mostly Cloud and Big Data based database
  - Size of data is more important than speed of data handling
  - Choose according to your situation (can be a MySQL at the beginning)



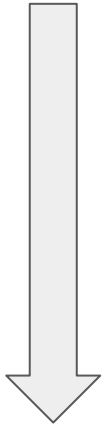
Data Warehouse: A separate database only for Data Stuffs

- Building a data warehouse is the first step to becoming a true data org
- Data Warehouse should be separate from the production database.
  - Dedicated database for leveraging data vs. production database
  - OLTP vs. OLAP (described in the slides later)



Data Infra: Foundation enabling data driven value generation

- Data Warehouse + Data Pipeline
  - The simplest form of Data Infra/Platform
- As the size of data increases
  - Data Lake is needed (more on this later)
  - Spark is needed as big data processing is required
- As data driven/informed decision matures
  - ELT solution is needed (“dbt”)
  - Dashboard solution is needed (Tableau, Superset, Power BI, …)
- As machine learning is being leveraged
  - ML related infra is required (NoSQL, API)
  - MLOps function is needed
  - Streaming data needs to be handled

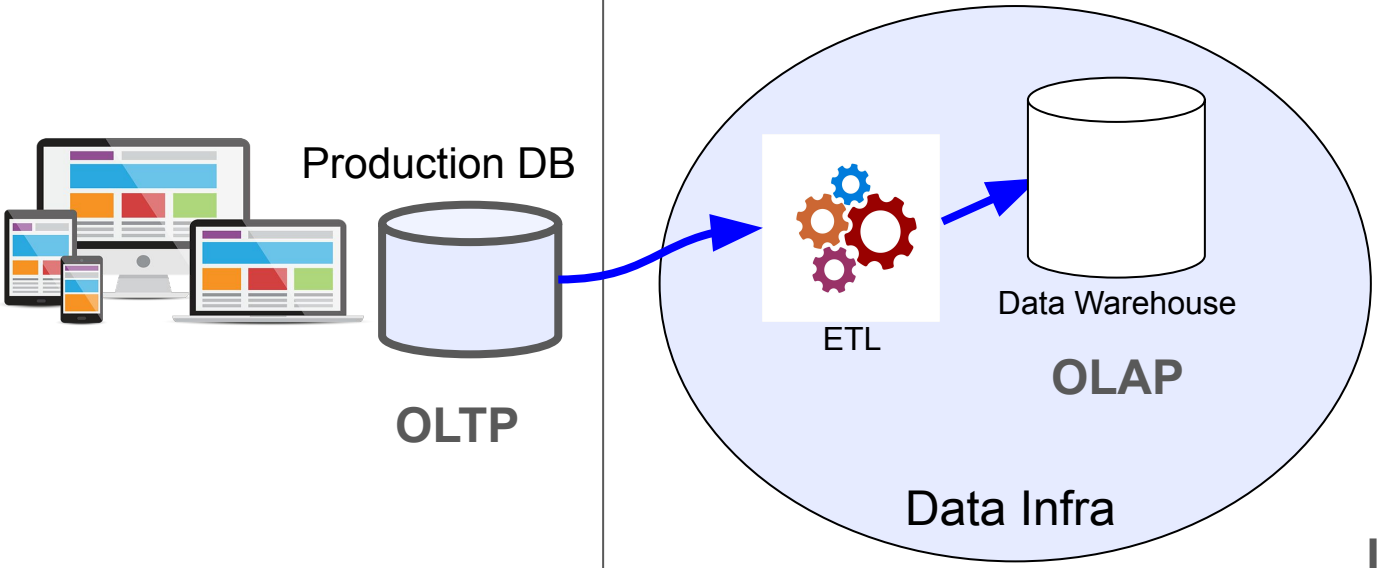


Sophistication of the infra

OLTP vs. OLAP

OLTP: OnLine Transaction Processing

OLAP: OnLine Analytical Processing



End Users, Customers

Backend Engineers

Data Engineers

Internal Audience

Difference between OLTP and OLAP

- OLTP: Production Database
  - Used to store data needed for operating the service/product
    - processing **speed** is more important
  - Managed by **Backend** engineering
- OLAP: Data Warehouse
  - Used to store data needed for data analysis and other data related work
    - **size of data** to handle is more critical
    - In other words, scalability is the key (and cost-effectiveness)
  - Managed by **Data** engineering

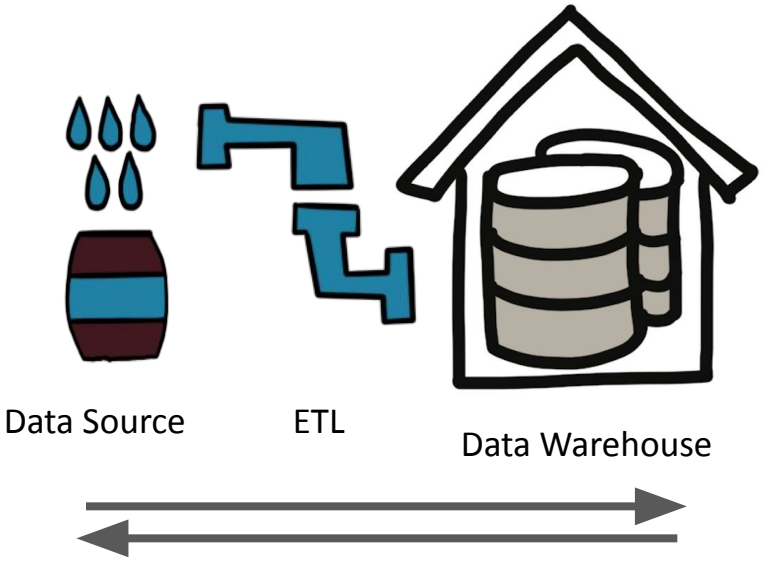
What are OLTP Databases and OLAP Databases?

- OLTP
  - MySQL, PostgreSQL, Oracle, …
  - A single server solution with limited capacity
- OLAP
  - Snowflake, BigQuery, Redshift, Hive/Presto, ClickHouse, …
  - A group of servers behind the scene

ETL vs. ELT

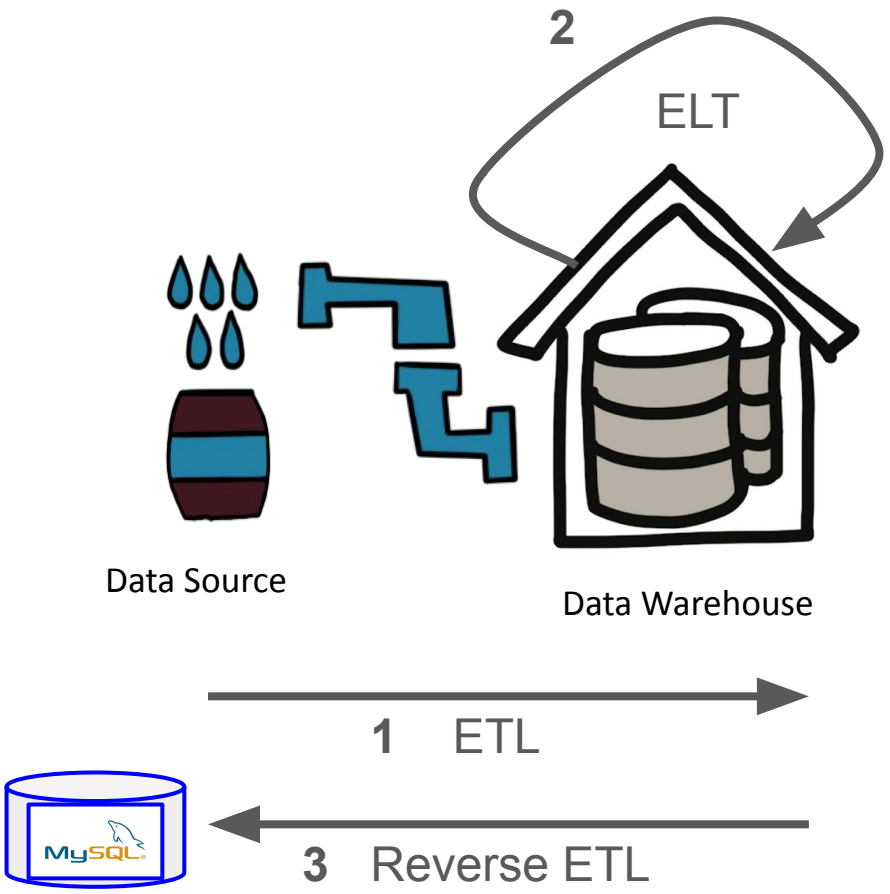
Definition of Data Pipelines

- ETL: Extract, Transform and Load
- Transport data from sources to destination
  - Destinations: mostly Data Warehouse or Data Lakes
  - Done via coding or SaaS (FiveTran or StitchData)
- Data Source Examples:
  - Production databases, log files, APIs, …
- Data Destination Examples:
  - Data Warehouse, Data Lake, …
  - Production Systems (Reverse ETL)



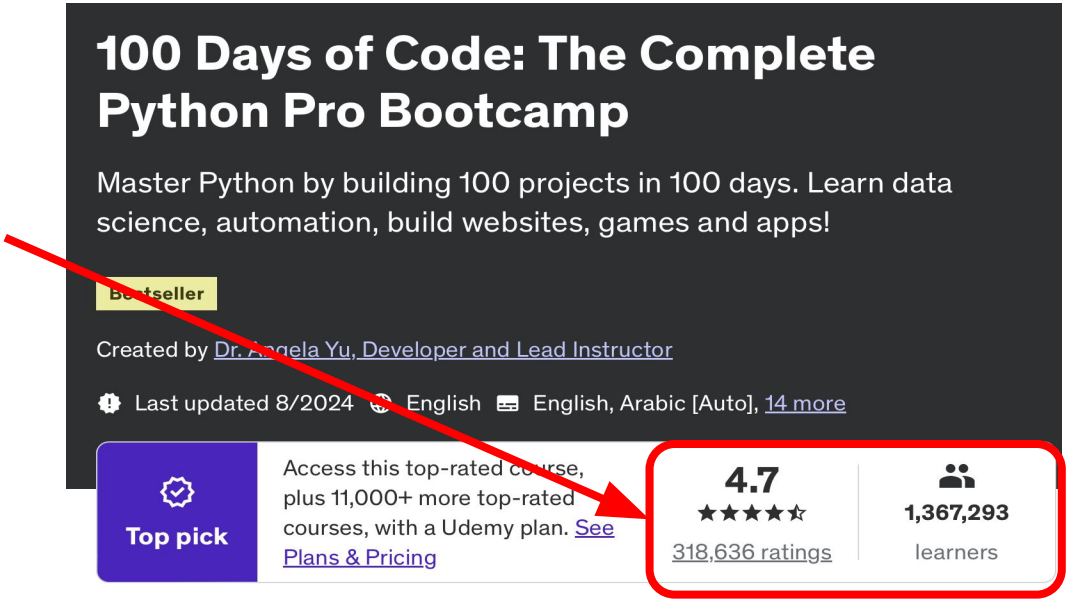
Types of Data Pipeline

1. ETL
2. ELT (“T”)
3. Reverse ETL



Reverse ETL example

1. Compute avg ratings, # of ratings and # of students per course in a data pipeline
2. Populate the results back to a production database, NoSQL and so on



ETL vs. ELT

- ETL
  - The process of bringing data from outside the data system to inside
    - Creates Raw Data
  - Done by data engineers
- ELT
  - The process of manipulating data inside the data system to create new data (usually more abstracted and cleaned)
    - Creates cleaner and more usable abstract info
  - The most popular tool is dbt: “Analytics Engineer”
  - Done by data analysts
  - Input to ELT can be in the data lake (more on data lake in the later slides)

Airflow Web UI Screenshot

Airflow | DAGs | Data Profiling | Browse | Admin | Docs | About

DAGs

Search:

		DAG	Schedule	Owner	Recent Tasks
	On	Branch_to_Redshift	15 11 * * *	john doe	2
	On	Build_Summary_Tables_Daily	25 13 * * *	john doe	6
	On	Build_Summary_Tables_Hourly	15,45 * * * *	john doe	10
	On	Cleanup_Logdir	0 12,20,0 * * *	john doe	2
	On	Download_Gsheet	30 14 * * *	john doe	1
	On	Export_Amplitude_To_Redshift	30 12,20 * * *	john doe	1
	On	Export_Salesforce_To_Redshift	30 11 * * *	john doe	2
	On	Postgres_to_Redshift_Daily	30 12 * * *	john doe	4
	On	Postgres_to_Redshift_Hourly	5,35 * * * *	john doe	40
	On	QuickBooks_to_Redshift	25 12,0 * * *	john doe	2

Data Warehouse vs. Data Lake

Weakness of Data Warehouse

- Not all data is structured (can’t be expressed as tables)
- What to do with unstructured data?
  - Unstructured data (ex: log files) tends to be much larger
  - Data Warehouse storage cost is too high to host unstructured data
- What if we can store unstructured data in a cheaper storage

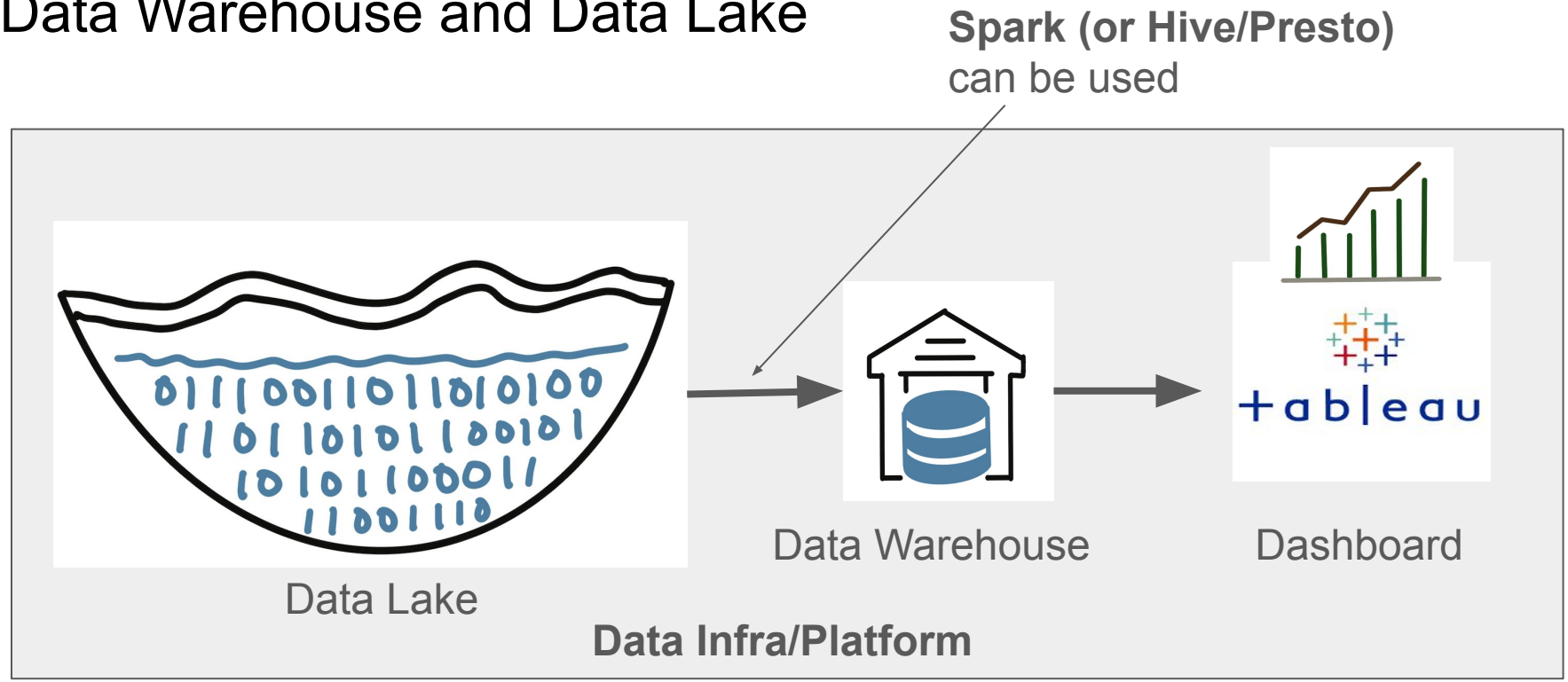
Data Lake to the rescue!

- Basically it is a very large file system (“Cloud Storage”)
  - In AWS, it would be S3 (Simple Storage Service)
  - In Google Cloud, it would be GCS (Google Cloud Storage)

Data Warehouse vs. Data Lake

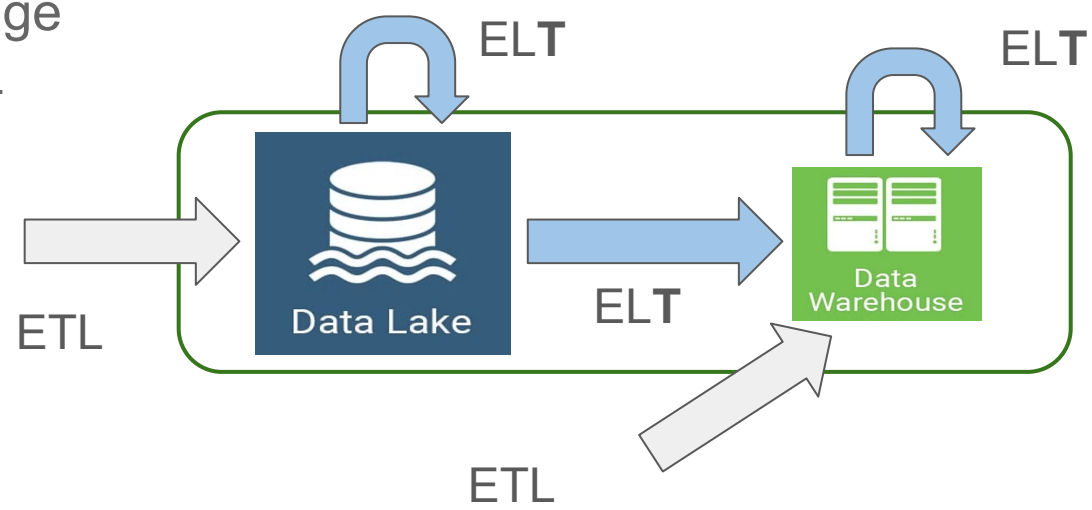
- Data Warehouse
  - Structured data is stored with some retention policy
  - Limited capacity with higher cost
- Data Lake
  - Structured data + Unstructured data
  - Order of magnitude bigger than DW
  - Cheap and big enough to store all the **raw** data without retention policy
- Both will likely be used together at some point
- LakeHouse
  - Data Lake + Data Warehouse
  - Iceberg from Netflix

Data Warehouse and Data Lake



Data Lake

- Order of magnitude bigger than Data Warehouse
  - Can store both **structured** data and **unstructured** data in cost effective fashion
  - Data can be stored in raw form without retention policy (as long as there are no legal issues)
- More like a cloud storage
  - AWS S3, GCP GCS, …



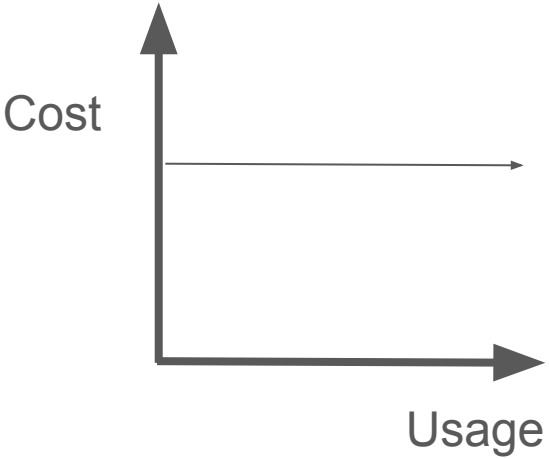
Break

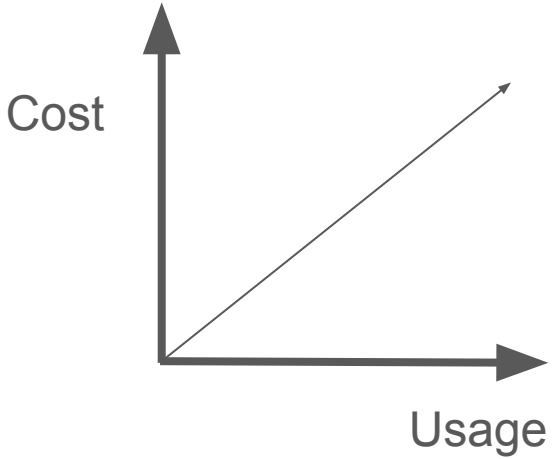
Data Warehouse Options

Data Warehouse: Different Options

- **Cloud** vs. On-Premise
- Open Source vs. Proprietary
  - Open Source ones are serviced from Cloud as well
- Fixed Cost vs. Variable Cost
  - CapEx (Capital expenditure) vs. OPEX (Operating expenses)

Fixed Cost vs. Variable cost





- Pay a fixed amount monthly for a predetermined capacity
- Ex) AWS Redshift dc2.large (2 CPU, 15GB memory, 160GB SSD): $180/month

- Pay as you go (true Cloud fashion)
- More scalable option and easier to manage but the cost can go too high
- Separation of Storage and Compute

What are the options

- AWS Redshift: Cloud, Proprietary, Fixed/Variable
- **Snowflake**: Cloud, Proprietary, Hybrid
- Google Cloud BigQuery: Cloud, Proprietary, Variable
- Apache Hive: Cloud/On-Premise, Open Source, Fixed
  - Can work with Data Lake
- Apache Presto: Cloud/On-Premise, Open Source, Fixed
  - Can work with Data Lake
- Apache Iceberg: Cloud/On-Premise, Open Source, Fixed/Variable
  - This is a Data LakeHouse solution gaining momentum

Basically all of these are big data based Data Warehouse/Lake

AWS Redshift

- AWS data warehouse started in 2012
  - Capable of distributed processing of PB-scale data
  - Supports PostgreSQL compatible SQL
  - Initially started with a fixed-cost model, but now also supports a variable-cost model (Redshift Serverless)
- Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Easy integration with other services within AWS
  - S3, DynamoDB, SageMaker, etc.
  - Supports execution of ML models (SageMaker)
  - Can be extended with services like Redshift Spectrum, AWS Athena and so on

Google Cloud Bigquery

- Google Cloud's data warehouse service (launched in 2010)
  - A flagship service of Google Cloud
  - BigQuery SQL is powerful and complex
  - Supports both variable and fixed cost options
- Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Easy integration with other services within Google Cloud
  - Cloud Storage, Dataflow, AutoML, etc.

Snowflake

- Started as a cloud-based data warehouse in 2014
  - Went public in 2020
  - Has now evolved to the point where it can be called a data cloud
  - Provides Data Sharing/Marketplace for selling data
  - Offers ETL and various data integration capabilities
- Enables big data storage, processing, and analysis based on SQL
  - Provides features for handling unstructured data and machine learning
  - Supports data formats such as CSV, JSON, Avro, Parquet, etc.
- Also supports S3, Google Cloud storage, and Azure Blob Storage

Sneak peek at Snowflake

Key Concept in Snowflake: Separation of Storage & Compute

- Snowflake separates storage from compute resources
- Compute resources are called "Virtual Warehouses"
- Virtual Warehouses can be scaled to different sizes based on computing power needs
  - We will be using X-Small by default

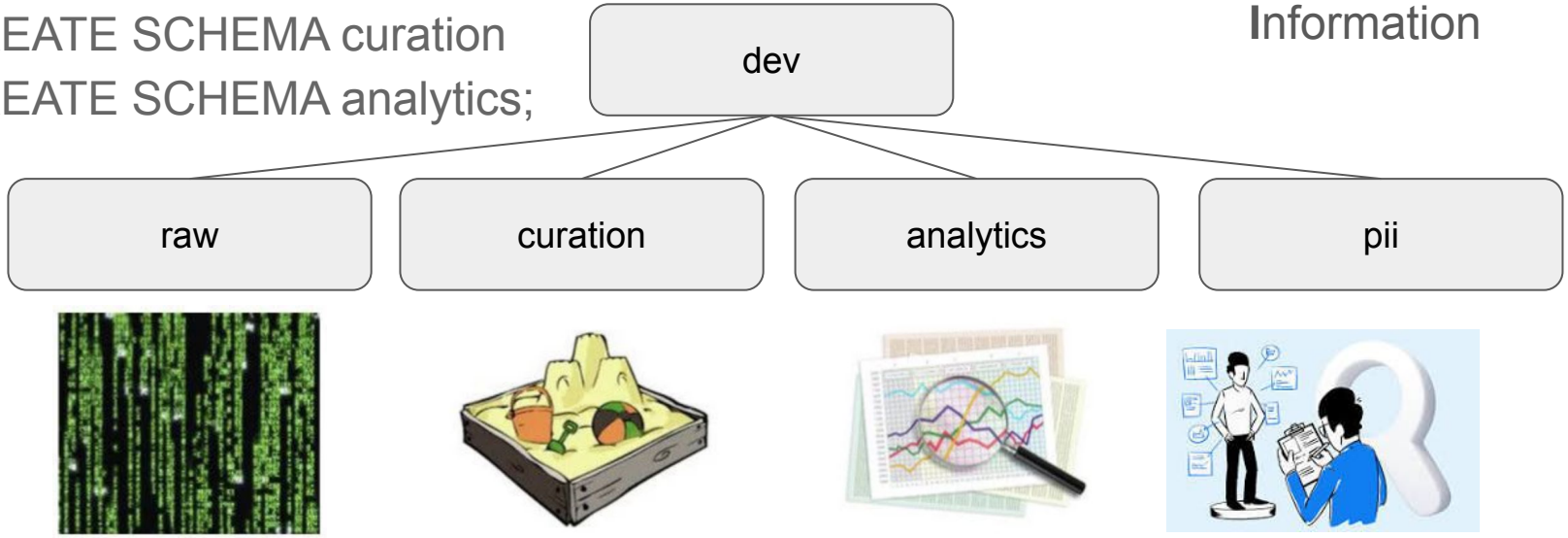
Key Concept in Snowflake: Database & Schema Organization

- Each student will be assigned their own dedicated database
  - For example, USER_DB_FOX
- Within each database, we'll organize tables using multiple schemas:
  - Raw schema: Contains source data and ETL (Extract, Transform, Load) tables
  - Analytics schema: Contains processed data and ELT (Extract, Load, Transform) tables

Database ➔ Schema: a way to organize your tables

CREATE DATABASE dev; (in our case USER_DB_FOX)
CREATE SCHEMA raw;
CREATE SCHEMA curation
CREATE SCHEMA analytics;

Personally
Identifiable
Information



ETL Tables

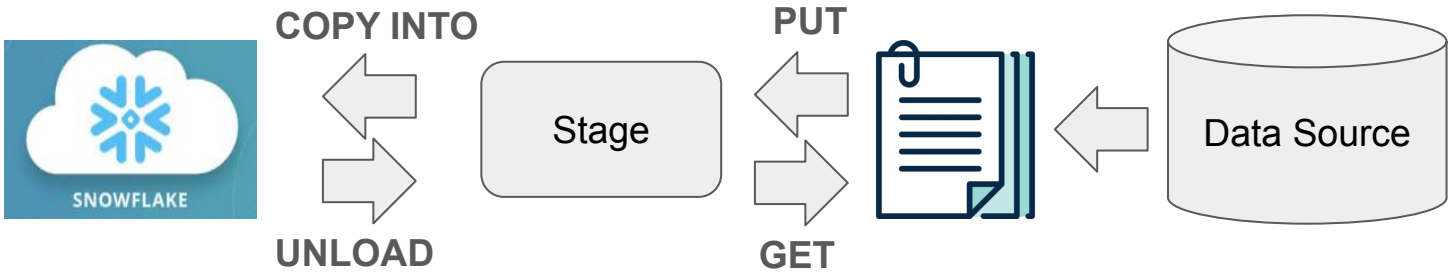
Cleanup

ELT Tables

PII Tables

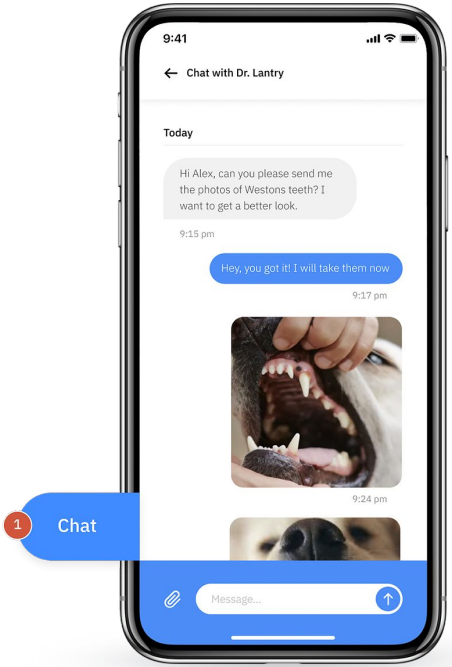
Key Concept in Snowflake: Bulk Operations

- Handling Large-Scale Data Uploads
  - INSERT INTO becomes impractical when working with millions of records
  - Snowflake provides the COPY INTO command for efficient bulk data loading
  - This approach significantly improves performance compared to INSERT INTO
- File-Based Processing with Stages
  - Snowflake processes data at the file level rather than record-by-record
  - An intermediary storage location called a "stage" facilitates this process
  - Stages act as a temporary buffer between your data source and Snowflake tables



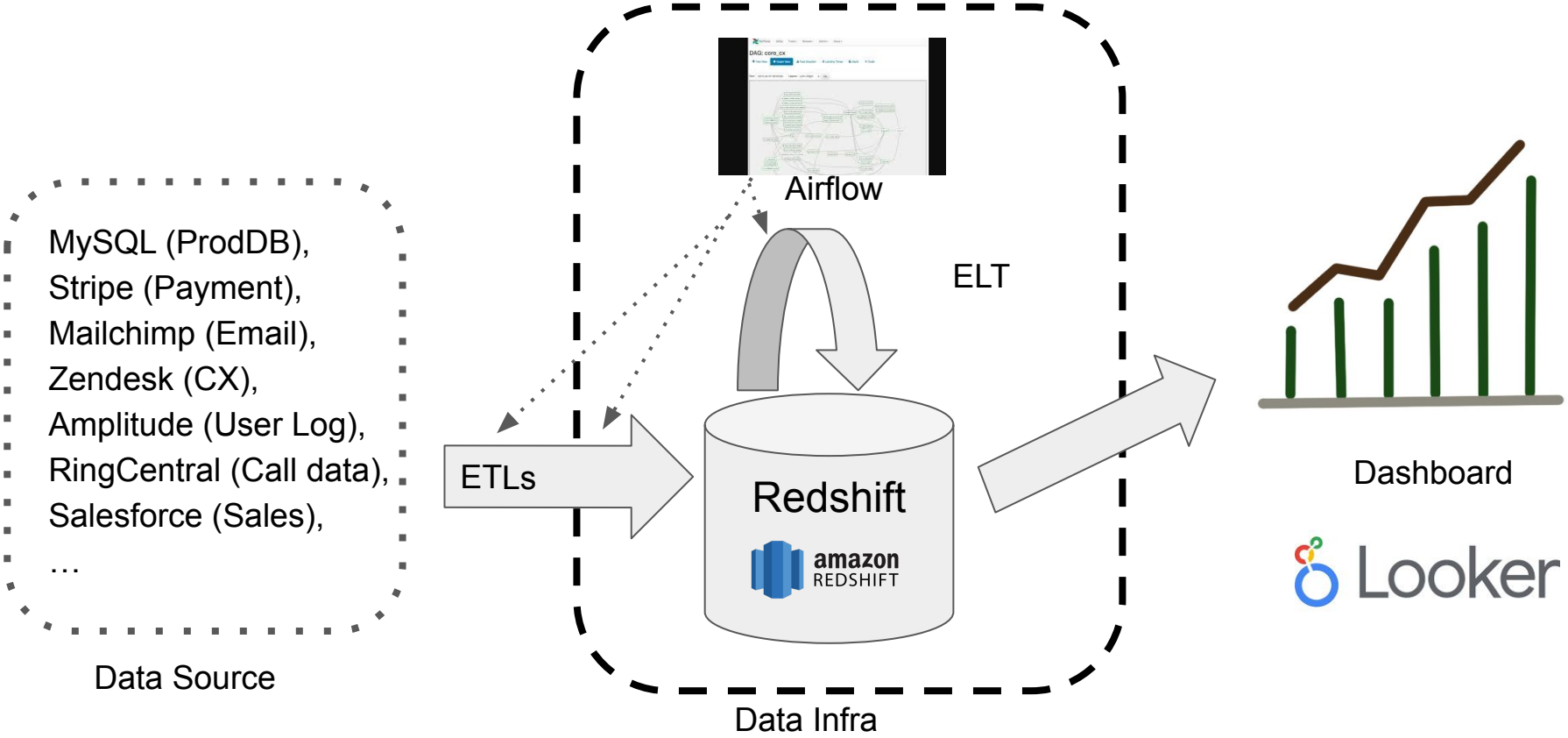
Example Data Platform

Case Study: Pet remote care marketplace service



- Matching Pet owners and Vets
- Components of its Data Infra
  - Airflow (Data Pipeline)
  - Redshift (Data Warehouse)
  - Looker (Dashboard)

Data Infra


MySQL (ProdDB),
Stripe (Payment),
Mailchimp (Email),
Zendesk (CX),
Amplitude (User Log),
RingCentral (Call data),
Salesforce (Sales),
…
Data Source
ETLs
Airflow
ELT
Redshift
amazon
REDSHIFT
Data Infra
Dashboard
Looker

Tables backed by ETL

From MySQL

- Vet
- Hospital
- Vet_Hospital
- Client
- Client_Vet
- Pet
- Pet_Client
- Case
- Payment
- Address
- Video Calls
- Reviews

From Amplitude

- Vet app
- Client app

From Salesforce

- Account

From Stripe (Credit Card Payment)

- Charge/PaymentIntent
- Refund

Tables generated by ELT

- Payment Summary
- Case Summary
- Vet Summary
- Client Summary
- Hospital Summary
- Client Cohort Summary
- Vet Cohort Summary
- Hospital Cohort Summary

- These are used in the dashboards and in the reporting
- **dbt** is the most popular choice
  - vs SQL CTAS (Create Table … As SELECT)

Lab 1

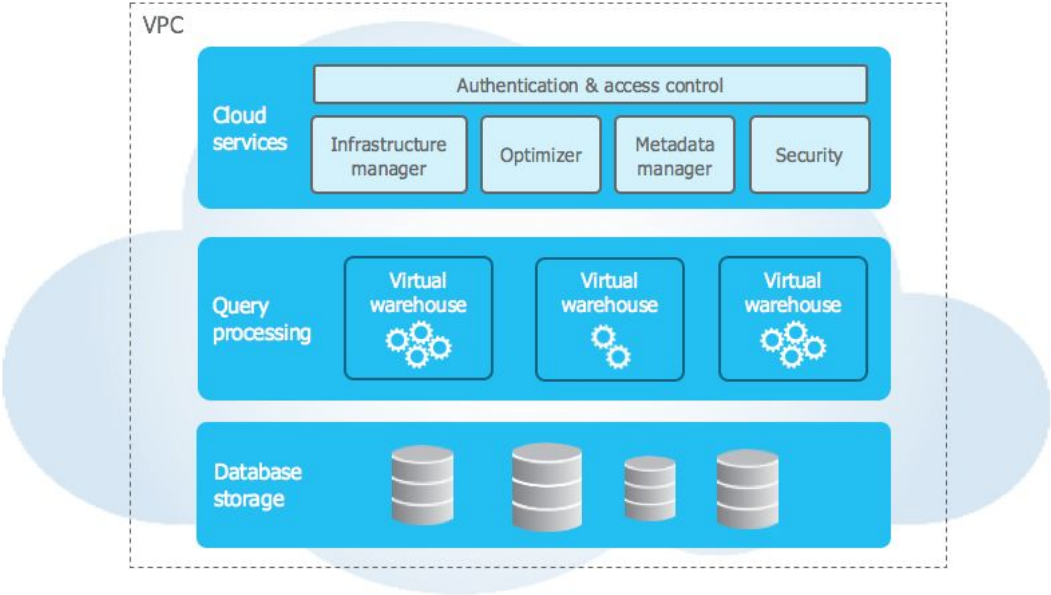
Lab 1

- “ETL Pipeline + Forecasting Pipeline on Snowflake”
- Team Selection due by next week
  - Section 24: https://sjsu.instructure.com/courses/1617421/groups#tab-110872
  - Section 23: https://sjsu.instructure.com/courses/1617436/groups#tab-110883
- More Details next Week

Demo: Snowflake Setup & Basics

Mini Recap on Snowflake

- Cloud-based data solution
- Separation of storage and compute
- SQL-based interface



https://docs.snowflake.com/en/user-guide/intro-key-concepts#snowflake-architecture

Homework

Homework #1

Assignment - Snowflake Environment Setup

Car Sales csv(synthetic) dataset is under the files/Homeworks/

1. (+1) Create Snowflake Account via Snowflake Academia as instructed
2. (+1) Create Schema 'car_sales_schema' under your Database. Use the warehouse of size XSMALL.
3. (+1) Create Stage 'car_sales_stage' in Snowflake and load the csv
4. (+2) Load Data into the 'car_sales_data' table (“COPY INTO” OR “UI LOAD”).
5. (+6) Write an SQL query to produce a report that shows the following insights
    1. The average number of cars sold per month in 2024, ordered Jan ➔ Dec
    2. The top 3 best-selling cars along with the count in 2024.
    3. The top 5 least-selling cars along with the count in 2024.

Submission Requirements:

- Submission should be a **PDF file.**
- Submission should comprise of Question, SQL Query and Snowflake Worksheet **Screenshot**.
- Snowflake Worksheet Screenshot should consist of query and results.

Q & A